

AI Ethics

Course Num: SEIS 651

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Week 2

Agenda

Focus Area

Fundamentals of AI

This is my favorite week because:

We're doing the data work!

Also, Datasheets for Datasets is my favorite paper.

Week 2

Outline – Key Points

- Lecture: Training Data
 - Sourcing
 - IP and Fair Use
 - Sampling
 - Preparation
 - Balancing
 - Data Workers
 - Synthetic Data
 - Governance (Lab Activity)

Lecture: Training Data

...And why it is so important

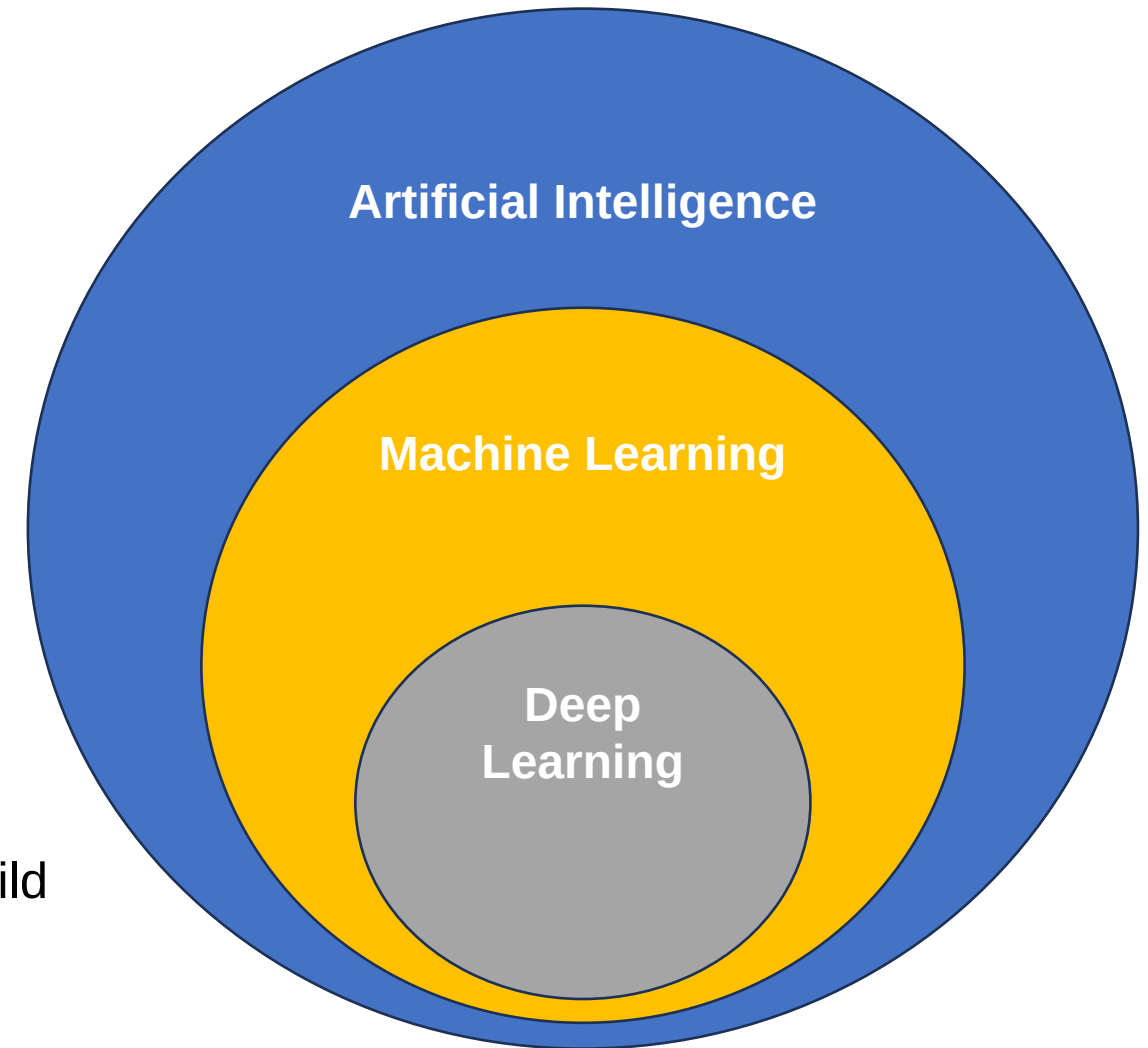
Recap - Week 1

- Defining AI
- Ethics across the AI lifecycle
- "Everybody wants to do the model work, nobody wants to do the data work."
- AI is not just "good programming" - you need training data!



Why Training Data?

- AI is built on data. We can't discuss AI ethics without discussing *data* ethics.
- Can I ignore this if my company just uses pre-trained models?
 - **No.**
- What if my company contracts out AI development? Can I ignore training data considerations?
 - **Also no.**
- **Why not?** One reason, among many:
 - "...a model is unlikely to perform well in the wild if its deployment context does not match its training or evaluation datasets, or if these datasets reflect unwanted societal biases." (Gebru et al., *Datasheets for Datasets*)

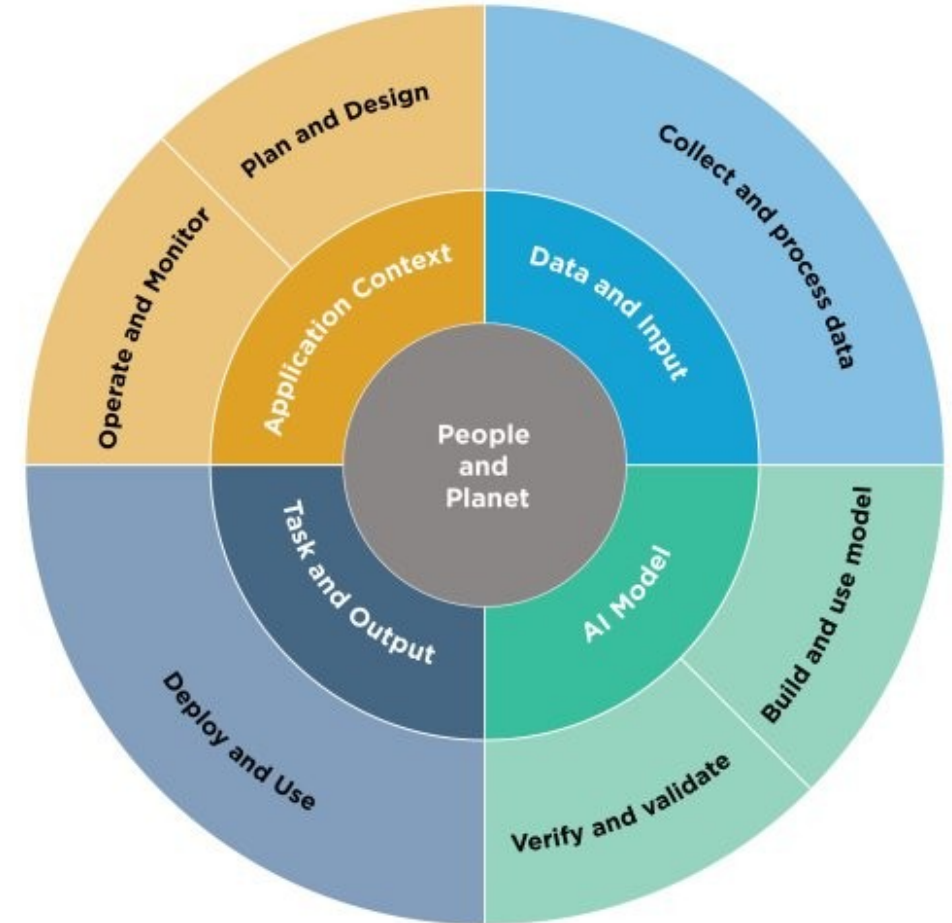


What is Training Data?

- Data used to "train" AI/ML models
 - Algorithms detect patterns in this data and make predictions from those patterns
 - Data is often split into training, testing, and sometimes validation sets
- Not all AI applications require you to collect/curate training data
 - Consider: Reinforcement learning?
 - Consider: Pre-trained models?
 - Key Takeaway: *you* may not have to collect the training data, but someone does!
- You're going to spend a lot of your time on the "data" work
 - You may need to evaluate whether *someone else* has sufficiently addressed the "data" work

AI Lifecycle

- Recall NIST's AI lifecycle – where does training data come in?
- The lifecycle of the **data** may differ from the lifecycle of the **AI product**!
- How much do you need, and when do you need it?
 - You need a lot.
 - Training isn't a "one time" event!



Sourcing

- Informed consent
 - Are you asking for permission or forgiveness?
- Data Quality vs. Quantity
- Context
 - RESEARCH your data! Was it collected for the purpose you're using it for?
- Does your training data represent your target population?
 - Also a very practical consideration!

Sampling

- Your training dataset should look like the population your model will represent
 - Sampling bias
 - Convenience sampling
 - Volunteer bias
 - Quality bias
 - “Specifically, we show that, without sampling bias, autophagy can lead to a rapid decline of both quality and diversity, whereas, with sampling bias, quality can be maintained but diversity degrades even more rapidly.” [Alemohammad et al.](#)
- How do you know when you have enough training data?

Sourcing and IP Concerns

Intellectual Property and Copyright

- Recent and developing cases: Getty Images, NYT vs OpenAI and Microsoft, Anthropic, and more!
- [Fair use](#)
- [Congressional Research Service on fair use and copyright](#)
- What is considered "publicly available" data?
- Law and Ethics
 - What about datasets or datapoints that aren't explicitly protected by law or regulation?
 - If you aren't a legal expert, you may need to pull in a legal expert.

Balancing and Data Prep

- What does it mean to "balance" data?
 - Under/oversampling, weighting, collecting more data, etc.
- What happens when your data is not sufficiently balanced?
 - Think about facial recognition...
- What happens when history or reality is unbalanced?
 - Job qualifications
 - Mortgage/loan approvals
 - Law enforcement
 - Fraud
- Should you exclude any data points from your training data set?
 - What about proxies?

Consider...

Balance

- Imbalance contributes to bias, inaccuracy, imprecision

Representation

- Models should represent the population they'll affect

Which is "better"?

- What's your use case?
- Are you describing or predicting?
- What are you expecting (or wanting) from your model's output?

Data Prep – Human Impact

Curating and Labeling Training Data

- Consider, for example...
 - How does a model learn to classify spam emails?
 - How does a language model learn to identify hate speech?
 - How does an image classifier learn to identify violent, explicit, or other graphic images?
- Who are the people behind this work?
 - <https://www.techimpactonworkers.iccr.org/6-exploitation-of-data-workers>
 - <https://cmr.berkeley.edu/2020/01/ghost-workers/>
- Consider projects like the BigScience ROOTS Corpus: <https://arxiv.org/pdf/2303.03915>

Synthetic Data

Synthetic Data

- Data created using generative AI or other means that don't involve collecting or measuring existing data points

Use Cases

- Balance datasets through augmentation (e.g., create more data by class, criteria, parameter, etc.)
- Create a representative dataset that would be too expensive, time consuming, or difficult to collect
- Structured or unstructured data

Synthetic Data

Pros

- Provides massive datasets that can be hard to come by in real life
 - Cheap, fast, relatively easy to gather
- Doesn't necessarily require a person to label data (e.g., for supervised learning)
- Not "human subjects" data (protects privacy, noninvasive)
- Can improve model accuracy vs. simply masking existing data or altering existing images (see [this article](#) from 2018, for example)
- Can help mitigate bias in some use cases

Cons

- Not always a perfect replacement
 - Remember, you often need to *have* data to make more data...
- Can result in less accurate model
- Can perpetuate bias in some use cases

Synthetic Data – Case Study

GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification

- Studied model performance by augmenting a relatively small dataset (182 images) with hundreds to thousands of synthetic images.
- “The classification performance using only classic data augmentation yielded 78.6% sensitivity and 88.4% specificity. By adding the synthetic data augmentation the results increased to 85.7% sensitivity and 92.4% specificity. We believe that this approach to synthetic data augmentation can generalize to other medical classification applications and thus support radiologists’ efforts to improve diagnosis.”

Synthetic Data – Case Study

Self-Consuming Generative Models Go MAD

- Studied retraining generative models with synthetic data. Three main focuses:
 - Fully synthetic loop
 - “...in this case either the quality (precision) or the diversity (recall) of the generative models decreases over generations.”
 - Synthetic augmentation loop
 - “...in this case fixed real training data only delays the inevitable degradation of the quality or diversity of the generative models over generations.”
 - Fresh data loop
 - “...in this case, with enough fresh real data, the quality and diversity of the generative models do not degrade over generations.”
- “Our primary conclusion across all scenarios is that without enough fresh real data in each generation of an autophagous loop, future generative models are doomed to have their quality (precision) or diversity (recall) progressively decrease. We term this condition Model Autophagy Disorder (MAD), making analogy to mad cow disease.” [Alemohammad et al.](#)

Retraining on Synthetic Data



Figure 18: Generation $t = 1$ of a **fully synthetic loop** with bias $\lambda = 1$. i.e., synthetic samples from [Alemohammad et al.](#) the first model $\mathcal{G}^1$.

Retraining on Synthetic Data



Figure 19: Generation $t = 3$ of a **fully synthetic loop** with bias $\lambda = 1$

[Alemohammad et al.](#)

Retraining on Synthetic Data



Figure 22: Generation $t = 9$ of a **fully synthetic loop** with bias $\lambda = 1$

[Alemohammad et al.](#)

Retraining on Synthetic Data



Figure 23: Generation $t = 1$ of a **fully synthetic loop** with bias $\lambda = 0.7$

[Alemohammad et al.](#)

Retraining on Synthetic Data



Figure 25: Generation $t = 5$ of a **fully synthetic loop** with bias $\lambda = 0.7$

[Alemohammad et al.](#)

Synthetic Data

Ethical considerations (including but not limited to...)

- Privacy
- Bias
- Representation

Practical considerations (including but not limited to...)

- Cost
- Accessibility/Availability
- Model accuracy

Recap – Week 2

- Training data is necessary for most AI applications
- Ethical sourcing (vs Legal?)
 - Remember the workers behind the data work
- RESEARCH and CRITICALLY EVALUATE your data (synthetic or not!)
 - Quantity and Quality
 - Balance vs Bias
 - Context
 - Governance
 - Etc.
- Training isn't a one-time consideration!

References

- [On The Ethics of Publicly Available Data](#)
- [What Is Publicly Available Data?](#)
- [U.S. Copyright Office – Fair Use and Copyright FAQ](#)
- [Stolen Stories or Fair Use? The New York Times v. OpenAI and the Limits of Machine Learning](#)
- [*Anthropic* and *Meta* Decisions on Fair Use](#)
- [Biased Programmers? Or Biased Data? A Field Experiment in Operationalizing AI Ethics](#)
- [Datasheets for Datasets Help ML Engineers Notice and Understand Ethical Issues in Training Data](#)
- [GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification](#)
- [Self Consuming Generative Models Go MAD](#)
- [Federal Data Strategy: Data Ethics Framework - https://resources.data.gov/assets/documents/fds-data-ethics-framework.pdf](https://resources.data.gov/assets/documents/fds-data-ethics-framework.pdf)

Class Activity

Datasheets for Datasets

Datasheets for Datasets

[Datasheets for Datasets](#) (Gebru et al., 2021)

Overview: What is a Datasheet?

- Purpose: provide a “...standardized process for documenting machine learning datasets,” not intended to be automated
- Key Stakeholders: dataset creators and dataset consumers
- Key Questions: “...motivation, composition, collection process, preprocessing/cleaning/labeling, uses, distribution, and maintenance.”

Application in Practice

- How do you think this practice would be received in your organizations?

Datasheets for Datasets

Apply tonight's material – could a datasheet address:

- Balancing
- Sourcing/Sampling
- Data Preparation Considerations
- Purpose/Use case
- Human Impact

Additional Governance Considerations:

- Privacy
- Don't forget sunseting!